

BLOCKCHAIN IN DIGITAL HEALTH AND LIFE SCIENCES

By: Daniel Burgwinkel and Richard Bergström

Summary: Blockchain brings the opportunity to improve processes throughout drug development, supply chain and health care delivery. The promising prospects of patient-reported outcomes, big data analytics and disease interception hinges on the integrity and authenticity of data. Fake data must be avoided as it can lead to wrong decisions, whether at the regulatory/payer level or in clinical practice. Society desperately needs new tools to manage data privacy, including next-generation tools for dynamic consent, with which citizens control access to their data. It has been demonstrated that cryptographic technologies using blockchain can be applied to life sciences and health care without creating new data silos.

Keywords: Blockchain, Digital Health, Cybersecurity, Data Sharing, Data Privacy

Introduction

The subject of blockchain is currently being discussed intensively by both business and information technology (IT) managers as well as regulatory authorities. The current state of play was well captured by the magazine *MIT Technology Review* in its May/June 2018 issue: the cover page read “Blockchain – hype or hope”. Everyone is trying to figure out where blockchain makes sense for them, and how to apply the new technology.

The Innovative Medicines Initiative (IMI), the world’s biggest public-private partnership in life sciences, recently launched a call on the use of blockchain in health care.^[1] The call text, drafted by pharmaceutical companies, cast a wide net ranging from supply chain to clinical research. Business managers see new disruptive business models, and the technology fascinates IT professionals

who have had the chance to collect preliminary experience from the digital currency Bitcoin. Furthermore, the Food and Drug Administration (FDA) is looking at blockchain, e.g. in the Information Exchange and Data Transformation (INFORMED) initiative: “...emerging technologies such as blockchain to enable secure exchange of health data at scale”.^[2]

What is Blockchain technology

The term blockchain describes a technological concept which stores data not in a central database, but rather distributed among the systems of users with the help of cryptographic protocols. The word “blockchain” was chosen because the data are stored in individual blocks which are then distributed and filed among the systems of the network participants and the order of the blocks is documented by means of a chain.^[3]

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Workshop 2: Blockchain for dummies

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Box 1: Distinguishing between different terms of Blockchain

Blockchain as a **technical concept** in IT, which uses cryptographic methods like hash-values and hash-trees that have been known for more than 30 years.

Blockchain **software** that provides programming code for cryptographic operations. In 2018, there are more than 50 different commercial as well as open source software products available.

Blockchain **applications** for the implementation of a certain use case. Typically, these applications are operated with the help of blockchain software or on a blockchain platform.

Blockchain **platforms** which use a certain software and are operated on the internet as a service, e.g. as an open peer-to-peer network or as a commercial service.

Blockchain as a service which provides the necessary software and services in a cloud. In such cases, a chosen blockchain solution can be operated on virtual computers in the cloud.

Even though it is only a technical concept, experts believe that this approach will revolutionise business models in various fields. If one wishes to use this technology for a given area of activity, the following questions arise:

- Which applications and use cases in Life Sciences can be realised on the basis of blockchains?
- What data can be sensibly stored in blockchains?
- Which transactions can be supported by blockchains?
- What technical restrictions exist?

A plethora of articles explaining the functional principles of blockchains can be found in the press and on the internet. For a basic understanding it is important to distinguish between the different terms (see Box 1). Depending on the concept,

Box 2: Key features of the Guardtime Blockchain for securing the eHealth System in Estonia

Which kind of data are stored in the Blockchain?

- Only cryptographic evidence is stored in the Guardtime KSI Blockchain.
- No patient data or personally identifiable information are stored in the KSI Blockchain.

How is patient data managed?

Patient data are stored in eHealth system and do not leave the borders of the Estonian systems. In the blockchain a cryptographic evidence (KSI signature) is created and stored in the blockchain and in addition in the eHealth application.^[5]

How do you proof data integrity in the overall process of patient data management?

Today typically every organisation runs a digital archive and there is no overall method to proof data integrity. With the new solution all involved health care organisations can check the integrity and provenance of the data with KSI Signatures.

How can you prove that you are compliant with patient consent?

In the current health care system, a patient has to trust that all parties act according to his consent. With the solution an evidence of data and process integrity in all processing steps can be created and matched with audit trails. In this way the research organisation can prove that the processing of the patient data was compliant with the patient consent.

business-relevant data can be stored in the blockchain (e.g. transaction data) or the data in the blockchain can contain references to external data, e.g. because the data requires a high storage volume or is confidential.

Blockchains can be employed in many areas and offer different functionalities. From a high-level view we can classify blockchain use cases in three categories:

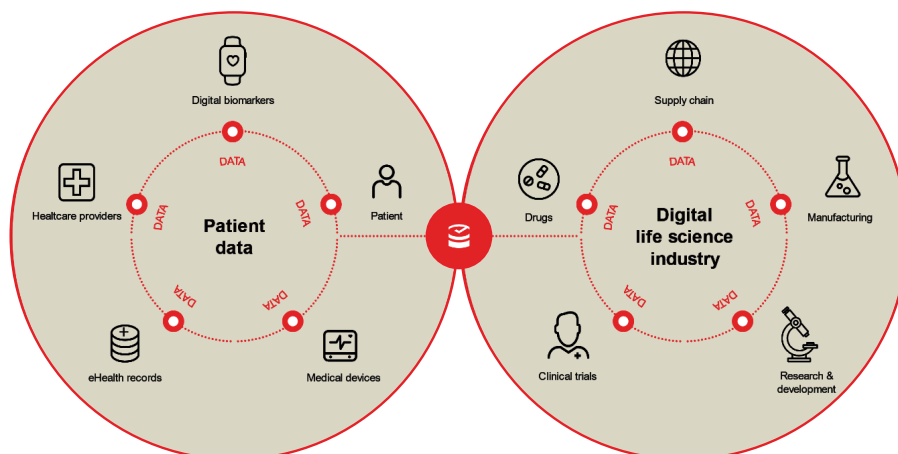
- Blockchains for proof of data integrity and for data sharing
- Blockchains for registration and certification
- Blockchains for the settlement of transactions

In this article we will focus on the first class of use cases, on data integrity and data sharing, that seems to have generated most of the interest so far in health care. Proof of data integrity can be provided using the blockchain, i.e. it can be verified that data has not been subsequently changed.

Practical example: Use of blockchains in eHealth in Estonia

In Estonia (see Box 2), blockchain technology is a component of the eHealth and eGovernment infrastructure and is used to ensure data integrity and to create an audit trail, which users have accessed the data.^[6] The following steps are carried out:

- The data, e.g. a new patient record, is generated in the respective application. The patient data are stored in the eHealth database and only the proof of integrity is transmitted to the blockchain.
- The eHealth infrastructure of Estonia is connected to the Guardtime blockchain service. The company Guardtime was founded in Estonia and operates a blockchain, which is used by government as well as industry customers.
- The eHealth data are checked for integrity at periodic intervals. The time interval depends on the need for protection. In the context of

Figure 1: Use Cases for Blockchain in Digital Life Sciences

Source: Guardtime

cybersecurity, for example, important data are checked at short intervals in order to detect any changes made to the data by an attacker. For long-term archiving, typically long time intervals are selected for monitoring. In addition to periodically checking the entire database, individual documents can be checked if necessary, e.g. an external auditor can check the authenticity of a document.

Estonia has realised the following advantages with this blockchain concept:

- In the eGovernment and eHealth infrastructure, a uniform procedure for guaranteeing data integrity and access logging has been implemented. The uniform procedure based on the blockchain saves costs and increases security, since not every authority or hospital has to implement its own procedure for data security and access logging.
- The citizen can access his/her data via a portal and trust the eHealth system that the data have not been manipulated and that access has only been granted to authorised persons.
- Since the data in eGovernment and eHealth are sensitive, and some data types have a large file size, e.g. medical image data, it does not make sense to store the data in the blockchain. Estonia has implemented a sensible concept here and stores only the proofs of integrity and not the data itself in the blockchain.

- The blockchain technology was introduced step by step and further expansion steps are planned. In the first step, the data integrity and cybersecurity described above were implemented. In further steps, the expansion for the processing of transactions, such as the use of digital prescriptions in health care, will take place.

Advantages of blockchain over current practice

Until now, similar functions have been implemented with the use of digital signatures or storage media with procedures for integrity protection. In comparison to using digital signatures or hardware-based integrity protection, a blockchain has the following advantages:

- Blockchain technology can prove both the integrity and the completeness of a set of data, as well as the chronological sequence. When using digital signatures, data are signed with the key of a person or organisation. Thus, management of public and private keys is necessary and can be very cumbersome in the long run. Blockchain technology is primarily a software-based procedure and is thus independent of the hardware used. So even for data stored in the cloud, data integrity can be verified.
- The use of a blockchain for proof of data integrity is based on the following procedure: data are generated outside of

the blockchain, e.g. a document or data set; the proof of integrity is generated by means of a hash algorithm and filed in the blockchain; data are checked for integrity at regular intervals. The time interval is chosen based on the protection requirement. Thus, in the context of cybersecurity, important data are checked at short intervals in order to detect any manipulation of the data by an attacker. For long-term archiving, longer intervals are typically chosen for monitoring. In addition to periodic checking of the entire data pool, individual documents can be checked as required, e.g. an external auditor can check the authenticity of a document.

These functions are of interest in application areas where proof that data hasn't been retrospectively manipulated is particularly important. Examples include research data for medical products, diagnoses in health care, or the configuration of machinery.

Potential for secure data management in research and development

The use of blockchain in the health system of Estonia shows how data integrity and security can be ensured with blockchain technology. In addition, blockchain technology will enable Life Sciences companies to transform their research and development process to ensure data integrity in all process steps and in all environments, for example when capturing data for digital biomarkers. **Figure 1** shows the use cases in the ecosystem of Digital Life Sciences.

Outlook

The outlook for blockchain is wide-reaching and important, including in the examples of clinical trials, sharing patient records, patient-reported outcomes and also in security. When running clinical trials, significant efforts go into source data verification. The emerging trend to run "virtual trials", with less involvement of a site with physicians and nurses, makes it even more important to care about data authenticity. For cross-border care, and even for elective care within one country, the sharing of health records provides a challenge: how to secure the safe sharing

of data, including the accompanying rules for protecting patient privacy. In the quest to focus on outcomes, and to pay for results, private and public players need to find ways to automatically capture data and keep track of the agreements they have made. Furthermore, the life sciences and health care sectors should be vigilant about cybersecurity threats which will likely increase in future.

References

- 1 Innovative Medicines Initiative (IMI). Available at: www.imi.europa.eu
- 2 FDA. Information Exchange and Data Transformation (INFORMED). Available at: <https://www.fda.gov/aboutfda/centersoffices/officeofmedicalproductsandtobacco/oce/ucm543768.htm>
- 3 Burgwinkel D. *Blockchain Technology*. Berlin, Boston: De Gruyter, 2016.
- 4 Republic of Estonia. e-estonia. Available at: <https://e-estonia.com/solutions/healthcare/>
- 5 Guardtime. Estonian eHealth Authority Partners with Guardtime to Accelerate Transparency and Auditability in Health Care, 2016. Available at: <https://guardtime.com/blog/estonian-ehealth-partners-guardtime-blockchain-based-transparency>
- 6 Guardtime. KSI Technology Stack. Available at: <https://guardtime.com/technology>
- 7 Khozin S. Digital Footprints in Drug Development: A Perspective from within the FDA. *Digital Biomarkers* 2017.

INCLUSIVE GROWTH AS A ROUTE TO TACKLING HEALTH INEQUALITIES

By: Emma Spencelayh

Summary: There are entrenched health inequalities between and within countries that continue to be unacceptable, in part because inequalities represent a failure of government to maximise the social and economic potential of its population. The Sustainable Development Goals provide a framework for cross-sector action and the translation of benefits from one sector to another. There are great opportunities for health policy makers and practitioners to tap into economic development agendas such as inclusive growth as a route to tackling health inequalities.

Keywords: Inclusive Growth, Health Inequalities, Social Determinants, Sustainable Development Goals

Revisiting the Declaration of Alma-Ata declaration

Thirty years ago, the declaration of Alma-Ata made it clear that the level of health inequality was ‘politically, socially, and economically unacceptable’.¹ Three decades on, we are still facing the issue of entrenched health inequalities between and within countries. For example, across countries, life expectancy varies by 34 years – a child born in Sierra Leone can expect to live for 50 years while a child born in Japan can expect to live for 84 years.² As an example of in-country inequalities, across the United Kingdom from 2014 to 2016, the gap in healthy life expectancy at birth between local areas with highest and lowest average health life expectancy was 18.4 years for females and 15.6 years for males.³

While there are clear and sound arguments for investing in health care systems as a route to improved health outcomes, tackling unacceptable variations cannot be left to the health care system alone, which is only one of many factors contributing to overall health outcomes. A wide variety of factors contribute to a person’s health and wellbeing, including access to education and good work, environmental factors such as decent homes and pleasant surroundings and strong social networks. These influences (the social determinants of health) are not distributed equally and are strongly shaped by government policy, including economic, social, housing and planning policies.

It can be difficult to determine the precise role the social determinants have compared to health care delivery or

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Economic strategies for health inequalities

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